Watermelon Sugar High: Identification of structural variants affecting fruit quality traits between wild and cultivated watermelons

Abstract:

Watermelon, *Citrullus lanatus*, is among the top five most consumed fruits globally. Its domestication has led to significant changes of fruit quality traits, including higher sugar content compared to its wild relatives such as *Citrullus amarus*. Genomic structural variants (SVs) have been reported to contribute to domestication traits. However, our knowledge about SVs between cultivated and wild watermelons and their phenotypic effects remains incomplete. To address this, a comparative study was conducted between a representative accession of *C. lanatus* subsp. *vulgaris* and a representative accession of *C. amarus*. Alignment of the high-quality genomes of these two accessions identified 111,738 SVs larger than 20 bp, which affected 19,482 and 19,096 genes in the cultivated and wild genomes, respectively. We found that 1,107 in the cultivated genome and 838 genes in the wild carried SVs affecting coding sequences, among which 25 had annotated functions related to disease resistance. RNA-seq data from cultivar and wild fruit flesh tissues were used for gene differential expression analysis. As a result, 2,732 genes in the cultivated genome and 4,884 genes in the wild genomes were differentially expressed during the fruit development. Through integrating SV and differential gene expression information, we identified candidate genes potentially crucial for fruit development and sugar content. This study has advanced our understanding of SVs and genes potentially affecting fruit quality traits and disease resistance. The SVs identified here serve as useful resources to facilitate future watermelon breeding efforts.

Personal Statement:

My experience at Boyce Thompson Institute this summer revolves around the incredible people I met. From engaging in lab work throughout the day to having fun playing sand volleyball, this has been a memorable summer. I am so grateful for all the friendships I've made during this time, and I look forward to witnessing all the remarkable achievements my peers will accomplish in the future.

During my time at the Fei lab, I improved my bioinformatics skills and expanded my scientific knowledge and experience. I have a newfound appreciation for those practicing computer science and analyzing large data sets. This rewarding and challenging experience has amplified my desire to pursue a Ph.D. I am excited to continue my scientific career and put my new skills to use.